HEMALATHA.K , Core Electronics Engineer

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Career Objective

ECE(B.E.2023-2027) student with a strong interest in Core Electronics,Embedded Systems,Iot,and Communication Technologies.participated in national level hackathons with hands on exposure to sensors,microcontrollers,and hardware based projects.passionate about applying engineering concepts to real world problems and seeking core electronics role to enhance technical skills and contribute to innovative solutions

Education

Bachelor of Engineering (CGPA : 7.2) 2023 – 2027
Kongunadu College of Engineering and Technology. Trichy, Tamil Nadu

Higher Secondary (72%) 2022-2023
The Trinity Academy Matric Higher Secondary School. Namakkal,Tamil Nadu

ACHIEVEMENTS

Participant IMPACTX 3.0 24-Hour National Level Hackathon focused on the 17 Sustainable Development Goals (SDGs).

Participant CREISTA 30-Hour National Level Hackathon focused on Hardware and Software.

Technical Skills

Core Electronics: Digital Electronics,Analog Electronics
Embedded Systems: Microcontrollers,Embedded Systems Basics
Communication: Basic Communicatin Systems,Computer Networks
Tools&Technologies: Mat Lab,Arduino IDE,LT Spice

Internship Experience

Full stack developer intern, Skylena infotech Private Limited. June 2025-July 2025

- Worked on full-stack web application development using HTML,CSS, and JavaScript.
- Assisted in frontend and backend integration connecting UI components with backend logic.
- Debugged application issues and improved functionality through testing.
- Followed Git-based collaborative workflows in an industry development environment.

Projects

CHAT APPLICATION WITH END TO END ENCRYPTION.

- Develop a secured real-time chat application with end-to-end encryption to ensure user privacy and data protection.
- Implemented strong cryptographic techniques (AES/RSA) to encrypt messages at the sender side and decrypt only at the receiver side.
- Tools : VS Code,HTML,CSS,MySQL,GitHub,Node.js,Webcrypto API.
- Role : Security and Encryption developer (worked on public & private key generation)

DOS ATTACK DETECTION SYSTEM.

- Developed a machine learning-based system to detect and prevent Denial Of Service(DOS) attacks in network.
- Implemented real-time traffic monitoring and analysis using pre-processing and classification algorithm.Utilized features like packet rateand traffic patterns to identify abnormal activites.
- Tools : VS Code,Python,Pandas,Sapy,Git&GitHub
- Role: ML Model Developer (attack normal classification,accuracy testing)

Certifications

Python 3.4.3

- IIT Bombay, Sep 2025.

HTML Program

- IIT Bombay, Feb 2026

VALUE ADDED COURSE

- Embedded system with AI Automation (Nov 2025)
- Leet Code Java (Sep 2024)
- SAWIT. AI LEARNATHON Program, HCL Tech ,Oct 2024.

NPTEL CERTIFICATIONS.

- Computer Architecture (Nov 2024)
- Industrial Internet of Things (Nov 2025)
- Enclosure Design of Electronics Equipment (Oct 2026)

Extracurricular Activities

- Skilled in video editing, photo editing, and videography.
- Continuously explores emerging technologies, including AI tools, automation platforms, and developer utilities.